

**NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
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BCS-5C

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INTRODUCTION

In modern computing environments, cloud networks have become the backbone of data exchange, supporting billions of devices and applications. Understanding how data packets flow, how routing decisions are made, and how congestion affects network performance is essential for students, researchers, and network engineers.

This project, Cloud Network Simulator, aims to create an interactive and visual simulation tool that represents how data travels in a cloud-based network. The simulator will provide real-time packet flow animation, latency and throughput measurement, congestion control visualization, and customizable network topology design — all within a user-friendly web interface.

Through this simulator, users can observe dynamic network behavior and gain practical insights into the internal operations of routing, performance metrics, and congestion mechanisms in a cloud network.

MOTIVATION

Traditional network learning environments rely heavily on command-line tools (like Cisco Packet Tracer or GNS3), which are often complex for beginners. There is a clear need for a visual and interactive web-based simulator that demonstrates core networking concepts intuitively.

This project is motivated by the following objectives:

- To provide an educational and research-friendly platform for understanding cloud networking.
- To enable real-time visualization of packet routing and network behavior.
- To allow simulation of network congestion and routing decisions under varying conditions.
- To measure and demonstrate key performance metrics such as latency, throughput, and packet loss.
- To give users the ability to design and visualize different topologies interactively.

By bridging theoretical networking principles with hands-on simulation, this project will make learning and experimenting with cloud network behavior both accessible and engaging.

CORE FUNCTIONALITIES (TENTATIVE)

Module	Description	Key Features / Functionalities
a. Packet Flow and Routing Visualization	Simulates and visualizes real-time packet movement across the network.	● Real-time animation of packet flow between clients, routers, and cloud servers. ● Color-coded packets (e.g., Blue – Client 1, Green – Client 2). ● Implements two routing algorithms: ○ Shortest Path Routing – Packets follow minimum-hop paths. ○ Load-Balanced Routing – Packets dynamically routed based on network load.
b. Latency and Throughput Measurement	Monitors performance metrics of packet transmission.	● Tracks average latency (ms) for all delivered packets. ● Measures real-time throughput (Mbps) with live graphical metrics. ● Displays packet delivery and loss counts in real time.
c. Congestion Control Simulation	Simulates and visualizes congestion behavior under varying network loads.	● Displays link utilization with color-coded congestion levels: ○ Green – Normal Load ○ Yellow – Medium Load ○ Red – High Congestion ● Simulates packet loss when congestion exceeds thresholds. ● Real-time router and link load monitoring.
d. Network Topology Design and Visualization	Provides an interactive and dynamic graphical representation of the network.	● Visualizes network components: 2 Clients, 3 Routers, 1 Cloud Server. ● Connects components through adjustable-capacity links. ● Displays node load and link usage dynamically. ● Interactive Controls: ○ Start/Pause simulation ○ Reset network ○ Adjust packet rate ○ Switch routing algorithm.

TOOLS

Component	Technology/Tool Used	Purpose
Frontend	React.js	To build the interactive user interface
Visualization	HTML5 Canvas API	For rendering nodes, links, and packet animation
Styling	Tailwind CSS	To design a clean, modern, and responsive UI
Icons	lucide-react	For UI icons and visual enhancement
State Management	React Hooks (useState, useEffect, useRef)	For simulation logic and reactivity
Environment	Node.js + npm	For package management and local deployment

TECHNOQUES AND METHODOLOGIES

The simulator combines network theory with front-end animation techniques. The development will follow these phases:

1. Topology Design Phase

Define network nodes (clients, routers, server) and connections with latency and capacity values.

2. Routing Logic Phase

Implement routing algorithms (Shortest Path, Load Balanced) to determine packet routes.

3. Packet Simulation Phase

Generate packets periodically based on a configurable packet rate, animate their movement, and track their progress.

4. Performance Measurement Phase

Calculate metrics such as latency, throughput, congestion, and packet loss in real-time.

5. Visualization Phase

Display all parameters through animations, color indicators, and dashboards for easy interpretation.

6. Testing and Optimization Phase

Ensure that packet movement, metric updates, and congestion reactions behave realistically and efficiently.

EXPECTED OUTCOMES

- A fully interactive network simulation environment with real-time visualization of packet flow and routing behavior.
- Clear understanding of how different routing algorithms (Shortest Path vs Load-Balanced) impact performance and congestion.
- Accurate measurement and live display of latency, throughput, and packet loss under varying network conditions.
- Real-time congestion visualization, helping users observe how network load affects performance.
- Dynamic representation of node and link utilization to demonstrate network efficiency.
- An educational and experimental tool for analyzing packet routing, congestion control, and performance metrics in a simulated cloud environment.

CONCLUSION

The Cloud Network Simulation and Visualization project will serve as an innovative educational and research platform that bridges the gap between theory and practice in cloud networking. It offers a visual, measurable, and interactive way to understand packet flow, routing algorithms, and congestion dynamics.

By using modern web technologies such as React.js and Canvas, the simulator will run efficiently on any browser without requiring specialized hardware or software installations, making it highly accessible to students, educators, and professionals alike.

